

Introducing RAG

Video Link: [Intro to RAG](#)

Retrieval Augmented Generation (RAG) is a technique that helps you work with large documents that are too big to fit into a single prompt. Instead of cramming everything into one massive prompt, RAG breaks documents into chunks and only includes the most relevant pieces when answering questions.

The Problem with Large Documents

Imagine you have an 800-page financial document and want to ask Claude specific questions about it, like "What risk factors does this company have?" You need to get the relevant information from the document to Claude somehow, but there are limits to how much text you can include in a prompt.

Claude with large documents

- We might want to ask Claude questions about specific sections of a large document
 - "What risk factors does this company have?"
- It's a private document - Claude won't have any idea what is contained in it
- How can we get Claude to understand the contents?

ANTHROPIC

Financial Doc, 800+ pages

Corporation
Annual Financial Report - Fiscal Year 2024

1. Executive Summary

Corporation ("The Company") is pleased to present its Annual Financial Report for the fiscal year ended December 31, 2024. The Company has demonstrated strong financial performance across all key metrics, with revenue growth of 12.3% year-over-year, reaching \$1.24 billion. Operating income increased by 15.7% to \$342 million, while net income rose by 18.2% to \$287 million.

The Board of Directors has approved a quarterly dividend of \$0.45 per share, representing a 10% increase from the previous quarter. This dividend will be payable on March 15, 2025, to shareholders of record as of February 28, 2025.

The Company continues to invest in research and development, with R&D expenses increasing by 22.5% to \$156 million. These investments are focused on expanding our product portfolio and enhancing our technological capabilities to maintain our competitive edge in the market.

2. Management Discussion & Analysis

Overview of Business Operations

Corporation operates in three primary business segments: Consumer Products, Enterprise Solutions, and Digital Services. The Consumer Products segment, which accounts for 45% of total revenue, experienced growth of 9.8% year-over-year. The Enterprise Solutions segment, representing 35% of total revenue, grew by 14.2%. The Digital Services segment, which accounts for 20% of total revenue, demonstrated the strongest growth at 18.7%.

Market Conditions and Competitive Landscape

The global market for our products and services remained competitive throughout the fiscal year. We observed increased competition in the Consumer Products segment, particularly in emerging markets. However, our strategic focus on premium offerings and brand differentiation allowed us to maintain our market share and improve profit margins.

Risk Factors

The Company continues to monitor several risk factors that could impact future performance, including:

- Fluctuations in raw material costs and supply chain disruptions
- Changes in regulatory requirements across our operating regions
- Technological disruptions and cybersecurity threats
- Economic uncertainty in key markets
- Foreign exchange rate volatility

Future Outlook

Looking ahead to fiscal year 2025, we anticipate continued growth across all business segments. We expect revenue growth of 10-12%, driven by new product launches and expansion into new geographic markets. We plan to maintain our focus on operational efficiency,

Option 1: Include Everything in the Prompt

The first approach is straightforward - extract all text from the document and stuff it into your prompt along with the user's question. Your prompt might look like this:

```
Answer the user's question about the financial document.
```

```
<user_question>  
{user_question}  
</user_question>
```

```
<financial_document>
```

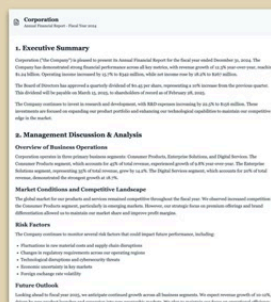
```
{financial_document}
</financial_document>
```

Possibly not the best solution

- There's a hard limit on how much text we can feed into Claude - the document might be too long
- Claude gets slightly less effective with very long prompts
- Larger prompts **cost more** to process
- Larger prompts **take longer** to process

ANTHROPIC

Option 1: Include the contents of the document in a prompt



Answer the user's question about the financial document.

```
<user_question>
{user_question}
</user_question>

<financial_document>
{financial_document}
</financial_document>
```

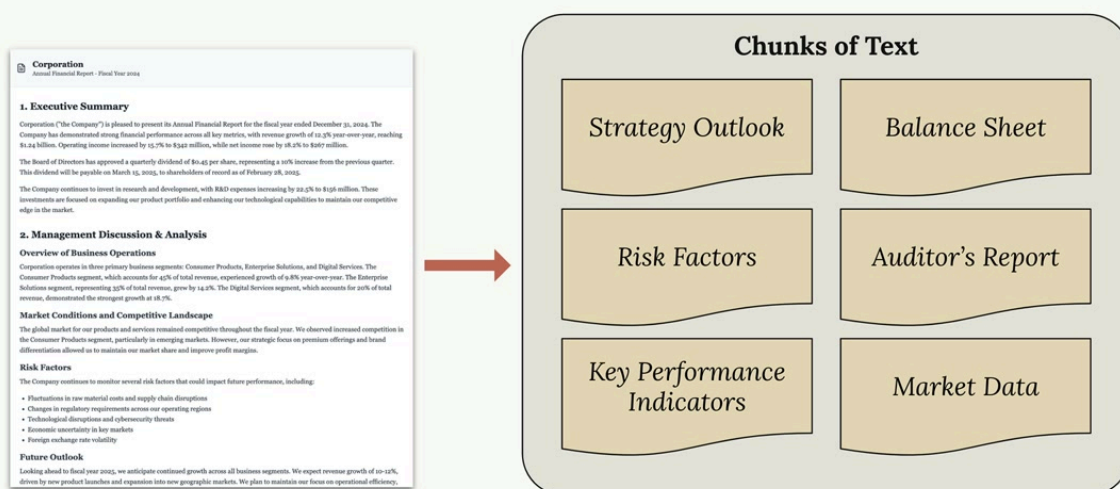
This approach has serious limitations:

- There's a hard limit on prompt length - your document might be too long
- Claude becomes less effective with very long prompts
- Larger prompts cost more to process
- Larger prompts take longer to process

Option 2: Break Documents into Chunks

RAG takes a smarter approach. First, you break the document into smaller chunks during a preprocessing step. Then, when a user asks a question, you find the chunks most relevant to their question and only include those in your prompt.

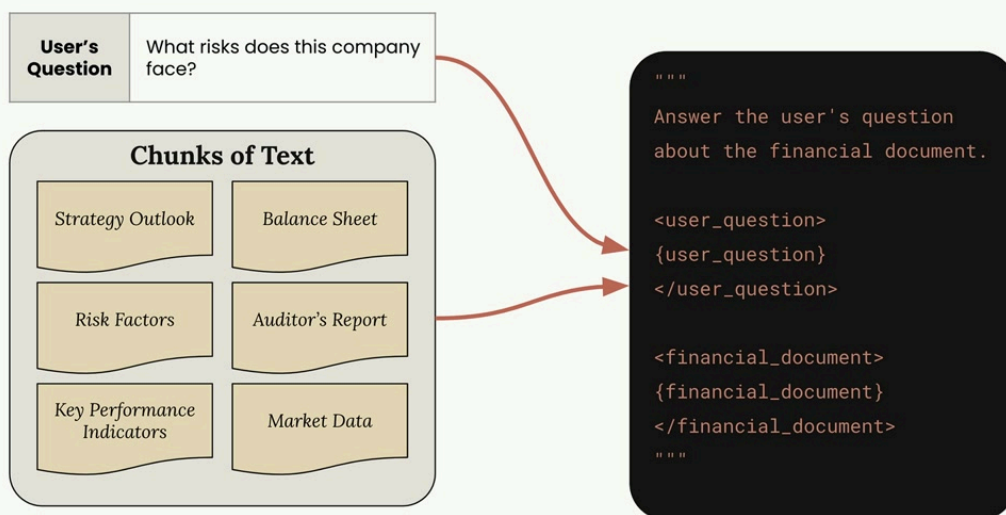
Option 2: Break the document up into many chunks. Put chunks relevant to the user's question in the prompt.



ANTHROPIC

Here's how it works: if someone asks "What risks does this company face?" you'd search through your chunks, find the "Risk Factors" section, and include just that relevant chunk in your prompt.

Option 2: Break the document up into many chunks. Put chunks relevant to the user's question in the prompt.



ANTHROPIC

Benefits of RAG

- Claude can focus on only the most relevant content
- Scales up to very large documents
- Works with multiple documents
- Smaller prompts cost less and run faster

Challenges with RAG

- Requires a preprocessing step to chunk documents
- Need a search mechanism to find "relevant" chunks
- Included chunks might not contain all the context Claude needs
- Many ways to chunk text - which approach is best?

For example, you could split documents into equal-sized portions, or you could create chunks based on document structure like headers and sections. Each approach has trade-offs you'll need to evaluate for your specific use case.

When to Use RAG

RAG involves many technical decisions and requires more work than simply including everything in a prompt. You'll need to analyze whether the benefits outweigh the complexity for your particular application. It's especially valuable when working with very large documents, multiple documents, or when you need to optimize for cost and performance.

The key insight is that RAG trades simplicity for scalability and efficiency. While it requires more upfront work to implement properly, it enables you to work with document collections that would be impossible to handle with simple prompt stuffing.